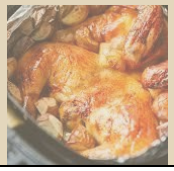
# Final Project

# Demand Estimation Lab



# Airfryer Market

An air fryer is a compact convention appliance used to heat up food through circulating hot air at high speeds, allowing it to fry food without using oil.

Within the following dataset, we will explore the air fryer market on Amazon to understand and infer markups and unit costs.

The top 10 brands are examined over 2019 to 2023 and notes the several features changes.

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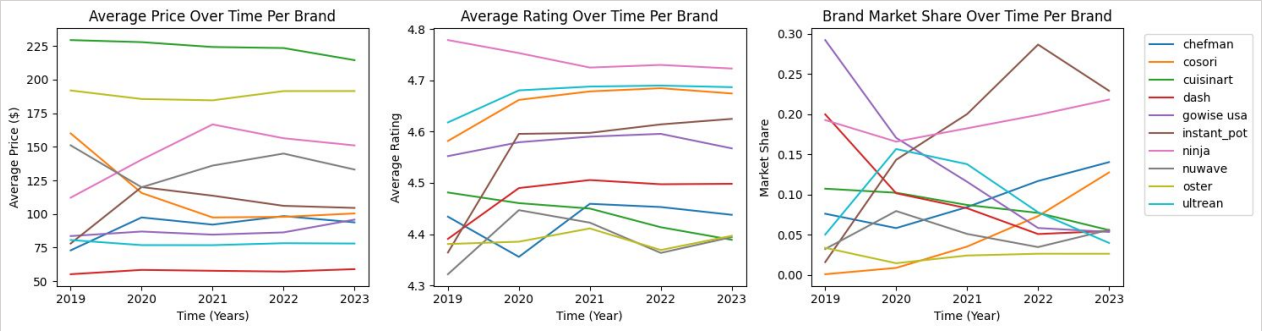
# Overtime...




Average Price
- Unstable brands include Ninja, Cosori, Instant Pot, Nuwave, and Chefman
- Stable brands include Cuisinart, Oster, Ultrean, Dash, and Gowise USA

Average Rating
- All brands are unstable in some respect

Market Share
- All brands are unstable except for Oster

We first examine the average price, average rate, and brand market share over time. It is important to see trends over recent years of these three characteristics because it illustrates the history of the brands popularity, and can potentially identify unintuitive trends. For example, the brand Dash consistently has the lowest average price of all the brands and is in the middle of the pack in terms of rating, however surprisingly, it's brand market share has decreased significantly over this time period. It will be important to determine possible causes for this decrease, either in the data or from other factors outside the set. But, is there a way to determine whether brand share explains characteristics of products in each brand?

# Demand Model

$$\log(s_{bt}) = \alpha_0 + \alpha_t + \gamma_b + \beta_{price} p_{bt} + \beta_{rating} r_{bt} + \sum_{\ell=1}^{L} \beta_\ell x_{bt\ell} + \epsilon_{bt}.$$

Goal – estimate a demand model where log brand share is explained by the feature characteristics price, rating, brand dummies, and year dummies

Model Type – multinomial logistic regression using linear regression package on log brand shares

Variables:
Dummy variables – dropped first brand (Chefman) and first year (2019) to use a baseline for comparison

X – combination of dummy variables, feature characteristics, and average rating and price

Y – Log brand share

We can explore this idea through a demand model where log brand share is explained by feature characteristics, price, rating, brand dummies, and year dummies. We can use a linearized multinomial logit model of the log brand shares and dummy variables to compare the estimated unit costs, markups, and price coefficient. These metrics show how strongly each brand is positioned in the market and how much pricing power they possess.

# Summary

## Estimates from Model

| | |
|---|---|
| Estimated Price Coefficient | -0.0377 |
| Estimated Unit Costs | 93.1022 |
| Estimated Markups | 29.7039 |
| Demand Slope | 0.0032 |

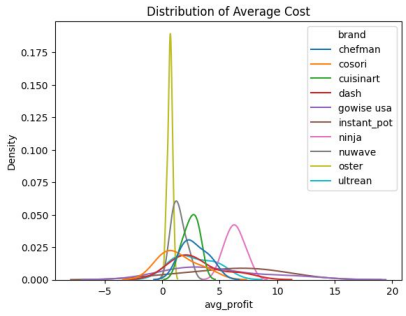
## Important Feature Coefficients

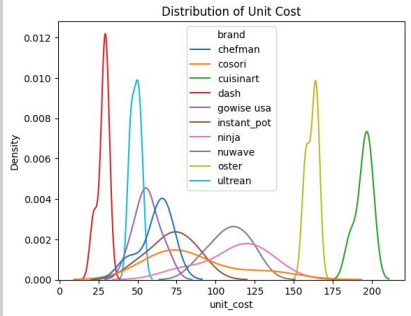
| | |
|---|---|
| Window Share | 12.8803 |
| Compact Share | 9.8153 |
| Brand Cuisinart | 6.4224 |
| Year 2020 | 0.1191 |

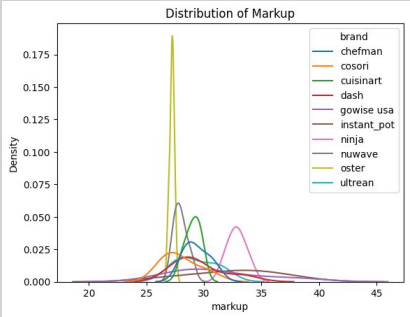
From the demand model, there were many figures that can help illustrate the position these brands have in the airfryer market. The following numbers summarize the airfryer market as a whole.

The price coefficient is an estimated effect of price on demand. We found a -0.0377 coefficient which means that higher prices reduces the brand share. The unit cost is the cost of producing one additional unit of a product. Therefore as we produce one additional product, the average cost is $93.10 more. The estimated markups is the percentage added to the cost of a product to determine it's selling price, and a 29% is very good for brands as they have more power over their pricing and selling. Demand slope is the derivative of the market share in respect to price, which in this case is a very slight, positive slope.

To the right shows the highest coefficients associated with the feature characteristics. These figures are an average of the difference between each product's figure and the baseline dummy variables that were dropped in the process of creating the model (Chefman and 2019)







A brand's relative position in the market is determined through analysis on it's price, feature characteristics, and its demand. Through the demand model and EDA, we are able to better understand how consumers interact with the products of each brand over the last 4 years. In the airfryer market, brand with high markups such as Gowise USA and Instant Pot have more flexibility to set their prices, where as brands with lower markups such as Oster and Nuwave face more competition pressures. Unit costs vary significantly and brands internally tend to have a higher density of a single price whether it's a higher or lower price.

# THANK YOU!